T7.2.3-Worksheet 1-Structure Diagram

Answer the following questions.

1. An **online cinema booking system** is an example of a computer system that is made up of sub-systems.

   The cinema booking system:

   - allows the user to browse available movies as posters or as a schedule list
   - allows the user to select their seats and pay for the booking
   - allows the cinema administrator to update movie timings and seat pricing.

   Complete the structure diagram for the given parts of the cinema booking system.

2. A **smart home security system** is an example of a computer system that is made up of sub-systems.

   The security system:

   - detects motion and checks for open doors or windows
   - sounds an alarm siren and sends an alert notification to the homeowner's smartphone
   - allows the homeowner to enter a PIN code to arm or disarm the system.

   Complete the structure diagram for the given parts of the smart home security system.

3. An **automated multi-storey car park system** is an example of a computer system that is made up of sub-systems.

   The car park system:

   - detects vehicle entry by scanning the car's number plate using an ANPR camera or issuing a paper ticket
   - displays the total number of currently available parking spaces on an external digital road sign
   - processes payment at the exit barrier by taking contactless card payment or calculating cash inserted.

   Complete the structure diagram for the given parts of the car park system.